236191: metagabbro, Bombora prospect

(Kurnalpi Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Kurnalpi (SH 51-10), Roe (3436)
MGA Zone 51, 458959E 6601401N

Sampled on 15 January 2019

This sample was collected from the 356.02 – 356.48 m depth interval of drillcore BBDD0007, drilled in 2016–17 by Breaker Resources NL at their Bombora prospect (Simson, 2017), with support from the Western Australian Government's Exploration Incentive Scheme (EIS). The drillhole is located on Yindi station, near the southwestern corner of Lake Roe, about 13.3 km east of Mount Charnleigh, 10.7 km west-southwest of Swan Lake, and 4.9 km north-northeast of May Day Tank.

Geological context

The unit sampled is a metagabbro (Fig. 1) of the Kurnalpi Terrane (Smithies, 1994; Groenewald et al., 2006). The unit is intruded into a 1.5 km-thick greenstone succession dominated by mafic meta-igneous and lesser metasedimentary and felsic meta-igneous rocks in the east–central Kurnalpi Terrane (Smithies, 1994; Simson, 2017). The succession is metamorphosed to upper greenschist to amphibolite facies, and includes a 400 m-thick fractionated dolerite sill, informally named the 'Bombora sill'. A porphyritic metadacite, sampled in outcrop about 14.4 km to the north, yielded an igneous crystallization age of 2673 ± 7 Ma (GSWA 112147, Nelson, 1995).

Petrographic description

The sample is a strongly deformed, recrystallized, medium- to coarse-grained quartz metagabbro (Fig. 1), consisting of about 35% amphibole, 25% plagioclase, 20% granophyre, 10% quartz, 2–3% magnetite, 2% phlogopite, 2% pyrrhotite, 2% carbonate minerals, 1–2% chlorite, <1% apatite and trace allanite. Amphibole consists mostly of highly irregular, elongate and crudely foliated masses of finely recrystallized grains but also includes regular tabular laths up to 4.5 mm long. Tabular laths and cores within the recrystallized masses are pale to olive green, and typical of hornblende, whereas the finely recrystallized masses are bluish green, consistent with more sodic compositions. It is probable that the amphibole was originally pyroxene, although relicts were not observed. Plagioclase is anhedral, equant to tabular, and up to 4 mm in size. Quartz-dominated granophyre forms irregular patches up to 10 mm in diameter, some partly overprinted by anhedral, strained, fine-grained quartz grains up to 0.5 mm in size. Magnetite is anhedral, equant, up to 2.5 mm in diameter, and exhibits a skeletal texture. Colourless to brown phlogopite forms subhedral laths up to 0.4 mm long in fibrous to radiating bundles. Pitted pyrrhotite stringers are narrow, up to 15 mm long, occur within the recrystallized amphibole aggregates, and include trace fine-grained, intergrown chalcopyrite and pyrite. Carbonate occurs along with quartz and minor chlorite in and around narrow veins. Apatite is euhedral, prismatic, elongate, and up to 1 mm long, and occurs mostly within recrystallized amphibole aggregates, together with minor, red–brown, unaltered, subhedral allanite up to 0.3 mm in size. The sample has been deformed and extensively recrystallized, probably at upper greenschist facies or slightly higher metamorphic grade.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque and anhedral to subhedral. The crystals are up to 300 µm long, and equant to elongate, with aspect ratios up to 4:1. Most zircons contain numerous inclusions of other minerals, are strongly altered and, in cathodoluminescence (CL) images, exhibit mottled textures, although relict concentric or broad planar zoning is preserved in many crystals. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 25–26 February 2021, using SHRIMP-B. Twelve analyses of the M257 standard were obtained during the session, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.18% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.22%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Nineteen analyses were obtained from 19 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

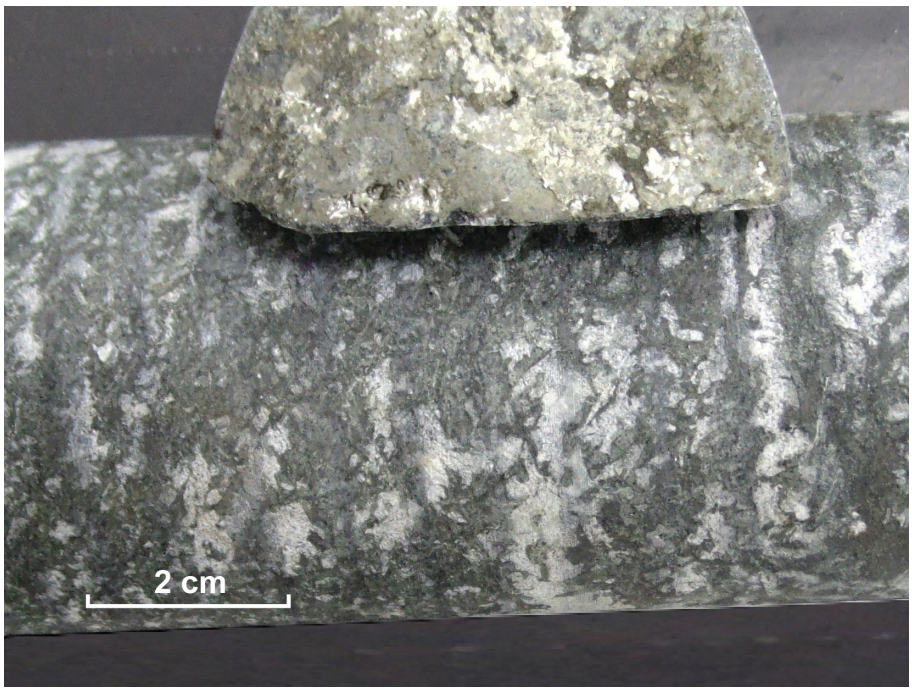

Figure 1. Drillcore image of sample 236191: metagabbro, Bombora prospect

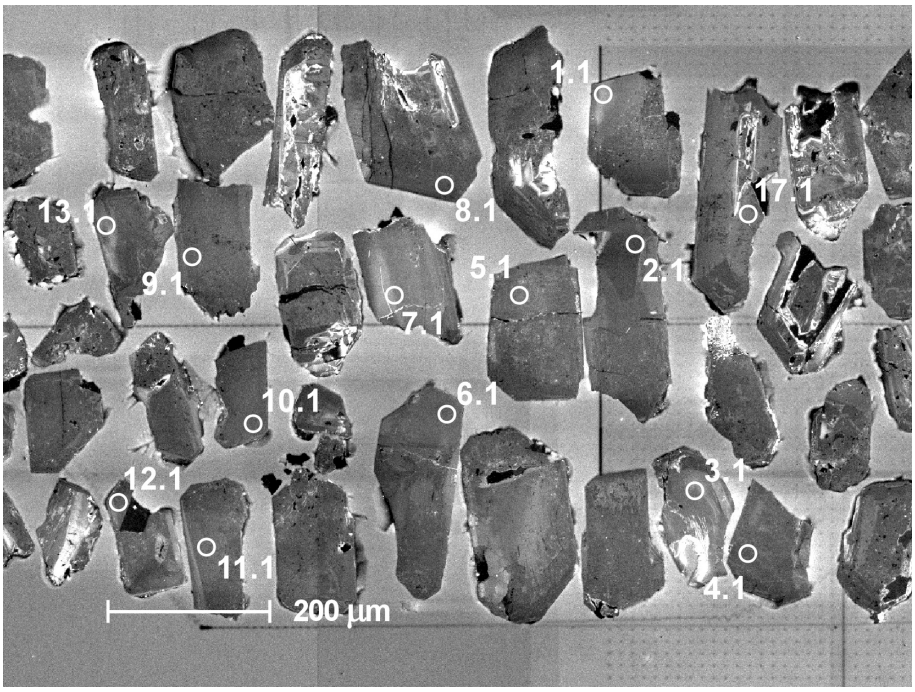

Figure 2. Cathodoluminescence image of representative zircons from sample 236191: metagabbro, Bombora prospect. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 236191: metagabbro, Bombora prospect

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	12	12.1	168	617	3.68	0.163	2.029	0.020	0.18452	0.00113	2.032	0.020	0.18307	0.00121	2580	21	2681	11	3.8
I	15	15.1	135	335	2.48	0.092	1.873	0.021	0.18530	0.00117	1.874	0.021	0.18448	0.00122	2756	25	2694	11	-2.3
I	8	8.1	290	2306	7.94	0.098	1.883	0.016	0.18542	0.00088	1.885	0.016	0.18453	0.00092	2744	19	2694	8	-1.8
I	17	17.1	112	116	1.03	0.036	1.917	0.022	0.18548	0.00125	1.918	0.022	0.18514	0.00127	2705	26	2700	11	-0.2
I	1	1.1	99	759	7.67	0.077	1.830	0.022	0.18609	0.00225	1.831	0.022	0.18539	0.00228	2809	27	2702	20	-4.0
I	19	19.1	110	461	4.21	0.020	1.927	0.023	0.18578	0.00132	1.927	0.023	0.18559	0.00134	2695	27	2703	12	0.3
I	4	4.1	343	2285	6.66	0.056	1.870	0.015	0.18637	0.00083	1.871	0.015	0.18587	0.00084	2760	19	2706	7	-2.0
I	3	3.1	80	500	6.26	0.051	1.882	0.024	0.18634	0.00148	1.883	0.024	0.18588	0.00151	2746	29	2706	13	-1.5
I	6	6.1	172	525	3.04	0.000	1.905	0.019	0.18595	0.00103	1.905	0.019	0.18595	0.00103	2720	22	2707	9	-0.5
I	13	13.1	101	315	3.11	0.000	1.897	0.023	0.18615	0.00136	1.897	0.023	0.18615	0.00136	2729	27	2708	12	-0.8
I	7	7.1	124	457	3.70	0.145	1.945	0.023	0.18752	0.00121	1.948	0.023	0.18622	0.00129	2672	27	2709	11	1.4
I	10	10.1	217	1120	5.16	0.018	1.890	0.018	0.18648	0.00096	1.891	0.018	0.18632	0.00097	2737	21	2710	9	-1.0
I	2	2.1	298	2060	6.91	0.006	1.823	0.015	0.18655	0.00084	1.823	0.015	0.18650	0.00084	2820	20	2712	7	-4.0
I	14	14.1	225	377	1.68	0.000	2.015	0.019	0.18657	0.00105	2.015	0.019	0.18657	0.00105	2598	20	2712	9	4.2
I	9	9.1	224	910	4.06	0.091	1.933	0.018	0.18747	0.00097	1.935	0.018	0.18666	0.00100	2686	20	2713	9	1.0
I	18	18.1	59	274	4.65	0.067	1.961	0.029	0.18753	0.00166	1.962	0.029	0.18693	0.00171	2655	32	2715	15	2.2
I	16	16.1	87	84	0.97	0.099	2.008	0.025	0.18839	0.00146	2.010	0.025	0.18751	0.00152	2604	27	2720	13	4.3
I	11	11.1	229	739	3.23	0.034	1.896	0.017	0.18938	0.00160	1.897	0.017	0.18908	0.00161	2730	20	2734	14	0.2
D	5	5.1	86	365	4.24	0.224	2.121	0.040	0.18260	0.00137	2.126	0.040	0.18059	0.00152	2486	39	2658	14	6.5

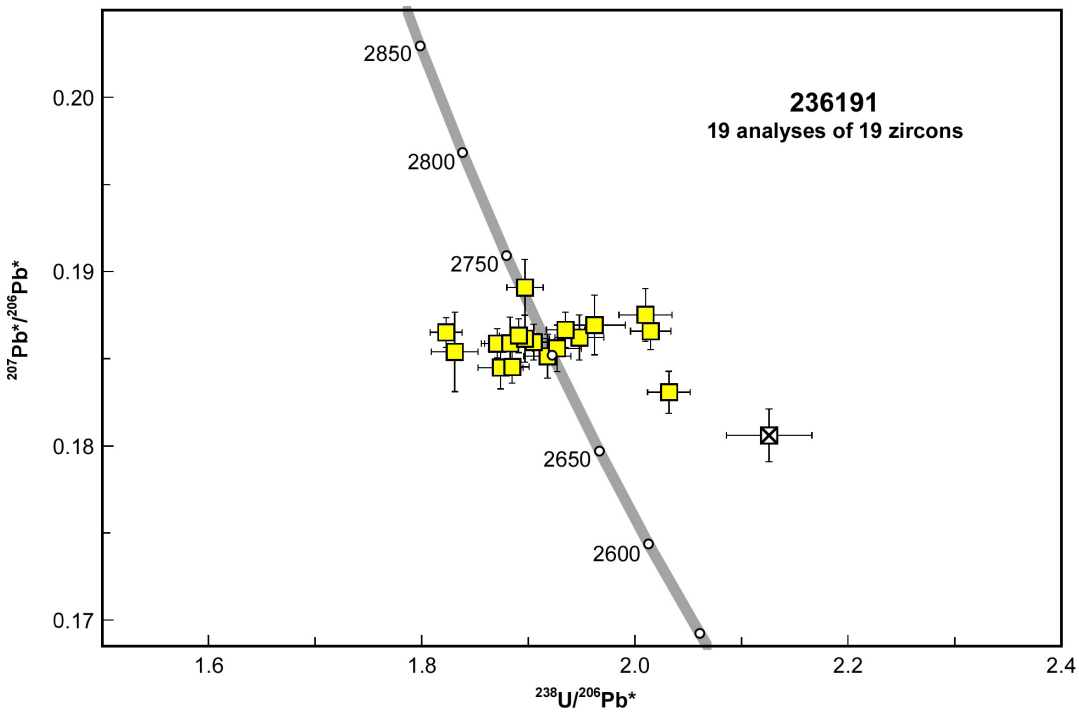

Figure 3. U–Pb analytical data for sample 236191: metagabbro, Bombora prospect. Yellow squares indicate Group I (magmatic zircons); crossed square indicates Group D (discordance >5%)

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). One analysis is >5% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable, and is considered not to be geologically significant. The remaining 18 analyses form a single group based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 18 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2706 ± 5 Ma (MSWD = 0.98). These analyses indicate mostly very high and variable Th contents (84–2306 ppm, median 500 ppm) and Th/U ratios (1.0 – 7.9, median 4.1).

The date of 2706 ± 5 Ma for the 18 analyses in Group I is interpreted as the age of igneous crystallization. Highly variable Th concentrations and Th/U ratios are consistent with the zircons having formed during igneous crystallization of the gabbro (e.g. Lanyon et al., 1993).

References

Groenewald, PB, Doyle, MG, Brown, SJA and Barnes, SJ 2006, Stratigraphy and physical volcanology of the Archean Kurnalpi Terrane, Yilgarn Craton — a field guide: Geological Survey of Western Australia, Record 2006/11, 25p.

Lanyon, R, Black, LP and Seitz, H 1993, U-Pb zircon dating of mafic dykes and its application to the Proterozoic geological history of the Vestfold Hills, East Antarctica: Contributions to Mineralogy and Petrology, v. 115, p. 184–203.

Nelson, DR 1995, 112147: porphyritic dacite, Yindi Woolshed; Geochronology Record 509: Geological Survey of Western Australia, 4p.

Simson, M 2017, Lake Roe Project Co-Funded Drilling Final Report (DAG2016-00840089): Breaker Resources NL, Exploration report A114837 (unpublished), 25p.

Smithies, RH 1994, Geology of the Roe 1:100,000 Sheet: Geological Survey of Western Australia, 1:100 000 Geological Series Explanatory Notes, 15p.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Recommended reference for this publication

Wingate, MTD, Lu, Y, Fielding, IOH and Smithies, RH 2022, 236191: metagabbro, Bombora prospect; Geochronology Record 1858: Geological Survey of Western Australia, 4p.

Data obtained: 26 February 2021

Data released: 29 April 2022